Missing inputs for the Section 10 ledger

2D fact, steady Euler, and the inputs the bootstrap does not have

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1 The 2D fact

A z-independent field has $\omega = \omega_3 e_3$, so ξ is constant and $|\sin\varphi| = 0 \le |y|/\rho$ for every $\rho > 0$. Constantin–Fefferman holds. 2D Navier–Stokes regularity is classical (Ladyzhenskaya). The transport equation $\partial_t\phi + u\cdot\nabla\phi = |\omega|^2$, $\phi(0) = 0$, still gives $\|\phi(t)\|_\infty \le \int_0^t M(s)^2\,ds$. On any interval where $M \ge m > 0$, the integral is at least $m^2 t$. Alignment does not delete the clock. Viscous 2D solutions decay, so this is not a claim that $\int_0^\infty M^2 = \infty$.

2 Steady Euler

For a steady Beltrami field, $\nu = 0$ and $M(t) \equiv M_0 > 0$, $\|\phi(t)\|_\infty = M_0^2 t$. The pasted driver then grows like $t^{3/2}$.

3 Inputs not supplied

Beale–Kato–Majda needs $\int M$. Metriplectic decay bounds $\int |\nabla u|^2$. Calderón–Zygmund puts ∇u in BMO, not L^∞. A peak of height M on a set of measure δ contributes order δM^6. Cosmic censorship is not a ZFC theorem about the unforced equations. Constantin–Fefferman is conditional. Statement (B) is not asserted.